

Indian Institute of Information Technology Guwahati

CS306 - Machine Learning, Mid-Semester Examination

Monsoon Semester AY 2025-26, Full Marks: 40, Time: 2 Hours

PART C : (Total Marks: 15)

Instructions: You have to write the answers in the answer script.

1. (5 points) Design a linear regression model using batch-mode gradient descent on the following dataset (two data points). Use this model to predict the output for the input pattern $[6 \ 2]$.

Dataset (two data points):

$$x^{(1)} = [1, 4], \quad y^{(1)} = 9$$

$$x^{(2)} = [2, 5], \quad y^{(2)} = 12$$

For demonstration, note the following:

- Learning rate: $\eta = 0.1$
- Number of training steps / epochs: 2
- All parameters are initialized as 1

Tasks (show step-by-step calculations):

- For epoch 1, compute $\hat{y}^{(i)}$ for both points, compute the batch gradients $\frac{\partial J}{\partial w_1}$, $\frac{\partial J}{\partial w_2}$, $\frac{\partial J}{\partial w_0}$, and update w_1, w_2, w_0 using gradient descent.
 - Repeat the computations for epoch 2 (using the parameters obtained after epoch 1): compute predictions, gradients, and parameter updates.
 - After epoch 2, compute the predicted output $\hat{y}$ for the input pattern $[3 \ 12]$ using the final parameters.
 - Briefly (in 1-2 lines) comment on whether the result after 2 epochs is likely to be close to the true value.
2. (5 points) Design a logistic regression model using mini-batch gradient descent (one epoch) on the following dataset (four data points). Use this model to predict the probability of the input pattern $[3 \ 1]$ belonging to class 1.

Dataset (four data points):

$$x^{(1)} = [1, 2], \quad y^{(1)} = 0$$

$$x^{(2)} = [2, 1], \quad y^{(2)} = 0$$

$$x^{(3)} = [3, 4], \quad y^{(3)} = 1$$

$$x^{(4)} = [4, 3], \quad y^{(4)} = 1$$

For demonstration, note the following:

- Learning rate: $\eta = 0.1$
- Number of epochs: 1
- Mini-batch size: 2
- All parameters are initialized as 1
- Loss: log-loss

Tasks (show step-by-step calculations):

- For Mini-Batch 1 ($x^{(1)}, x^{(2)}$), compute $\hat{y}^{(i)}$, the gradients $\frac{\partial J}{\partial w_1}$, $\frac{\partial J}{\partial w_2}$, $\frac{\partial J}{\partial w_0}$, and update parameters.
 - For Mini-Batch 2 ($x^{(3)}, x^{(4)}$), repeat the calculations using the updated parameters from Mini-Batch 1.
 - After completing one epoch, compute the predicted probability $\hat{y}$ for the input pattern $[3 \ 1]$ using the final parameters.
 - From the computations of the mini-batches, discuss briefly, if the gradient of the loss is increasing (or decreasing), how the parameter updates will be effected.
3. (5 points) Apply the K-means clustering algorithm on the following dataset of 4 points to form $K = 2$ clusters. Perform clustering for two full iterations (assignment + update). Show all calculations step by step.

Dataset (four data points):

$$x^{(1)} = (2, 3), \quad x^{(2)} = (3, 3), \quad x^{(3)} = (8, 7), \quad x^{(4)} = (9, 8)$$

For demonstration, note the following:

- Number of clusters: $K = 2$
- Initial centroids (given):

$$C_1 = (2, 3), \quad C_2 = (8, 7)$$

- Distance measure: Euclidean distance

$$d(x, c) = \sqrt{(x_1 - c_1)^2 + (x_2 - c_2)^2}$$

- Update rule: New centroid = mean of all points assigned to the cluster.

Tasks (show step-by-step calculations):

- (a) Compute the distance of each point from both centroids C_1 and C_2 for both the iterations.
- (b) Assign each point to the nearest centroid (cluster assignment step) for both the iterations.
- (c) Recompute the centroids C_1 and C_2 based on the new assignments (update step).
- (d) After one full iteration, list the new clusters and their updated centroids.